Robert Thijs Kozma

AI RESEARCH AND TECHNICAL LEADERSHIP

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AI researcher and mathematician building NLP, agentic, and multimodal systems. Combines hands-on R&D with technical direction and product development, leading work from research questions through prototyping, validation, and production handoff with product and engineering teams.

EXPERIENCE

Principal Research Data Scientist - iSpot, Bellevue Sept 2024 - Oct 2026

Senior Research Scientist, May 2022 - Aug 2024

- **SAGE AI:** Architected agentic workflows that turn natural-language questions into plans, generated code, data execution, and analytical results. Built the end-to-end prototype and advanced the architecture into production.
- **NLP inputs to SAGE:** Built hierarchical theme-discovery pipelines that supply SAGE with structured consumer-feedback data. Combined embeddings, clustering, and language models to organize themes across ads, brands, categories, and industries, with a taxonomy viewer for exploration.
- **Transformer training:** Trained a transformer model to classify emotional responses expressed in viewers' written feedback on advertising, turning free-form language into structured measures for Ace Metrix creative assessment.
- **Product development:** Developed multimodal prototypes combining video, audio, and text, and product proposals that helped secure executive backing for AI creative analysis.
- **Statistical measurement:** Developed census-derived survey weights and generalized demographic slicing; analyzed effects on creative scores and brought revised methods into production.
- **Architecture and delivery:** Defined component interfaces with engineering, stress-tested agent behavior, optimized model cost/performance, and integrated production model access through AWS Bedrock.

LEADERSHIP AND RESEARCH

Team Lead - University of Illinois Chicago May 2020 - Aug 2021

Led a **15-member team** providing course support to **150+ faculty** during remote instruction; delivered online orientation for **1,500 students**.

Mathematics Research - University of Illinois Chicago Ph.D. 2021

Proved statistical results for Markov-driven random matrix products and developed algorithms and packing-density bounds in hyperbolic geometry. Published in *Discrete & Computational Geometry*, *Geometriae Dedicata*, and *Ergodic Theory and Dynamical Systems*. Publications.

Machine Learning Research - Boston University and Missouri S&T 2008 - 2011

Developed fuzzy ART neural networks for neuromorphic hardware under DARPA SyNAPSE with HP Labs; coauthored work on memristive-fuzzy hardware and pdi-Boosting for brain-computer interfaces.

TECHNICAL SKILLS

Languages	Python, SQL, MATLAB, Mathematica, LaTeX
NLP and ML	BERT, transformers, PyTorch, TensorFlow/Keras, Hugging Face, scikit-learn, NumPy/Pandas
AI development	Claude Code, Kiro, coding-agent evaluation and tool integration
Cloud / AI	AWS, Amazon Bedrock, model APIs, GPU compute, inference serving, cost optimization

EDUCATION

University of Illinois Chicago - Ph.D., Mathematics 2021

Stony Brook University - M.A., Mathematics 2013

Boston University - B.A., Pure and Applied Mathematics 2010

Summa cum laude, with distinction.

Selected honors: U.S. Junior Oberwolfach Fellow; UNKP Doctoral Research Grant (2018).

Languages: English and Hungarian (native); Japanese (JLPT N1).

Hobbies: Skiing, cooking, and the outdoors.